

Comprehensive Analysis of ESG Disclosure and Sustainability Performance: IRPC Public Company Limited (2014–2023)

The corporate landscape of Southeast Asia has undergone a profound transformation over the last decade, transitioning from a focus on rapid industrial expansion to a more nuanced model of "Sustainability Excellence." Within this context, IRPC Public Company Limited (IRPC) stands as a significant case study of an integrated petroleum and petrochemical entity navigating the complexities of global Environmental, Social, and Governance (ESG) standards. This report provides an exhaustive evaluation of IRPC's ESG disclosure evolution, strategic alignment with international frameworks, and its performance trajectory from 2014 through 2023.

Company Identification and Strategic Business Context

IRPC Public Company Limited is the legal name of the entity identified on the Stock Exchange of Thailand (SET) by the ticker IRPC.BK.¹ The organization functions as a premier integrated petroleum and petrochemical producer, operating Southeast Asia's first fully integrated petrochemical complex located in Rayong Province, Thailand.² The headquarters are situated at 555/2, Energy Complex, Building B, 6th Floor, Vibhavadi Rangsit Road, Chatuchak, Bangkok 10900.¹

The primary business activities of the company are categorized into five distinct yet synergetic segments. First, the refinery business operates a facility in Rayong that processes crude oil into refined products, including diesel, gasoline, lube base oil, and fuel oil.¹ Second, the petrochemical business produces polymers and various chemical products, such as polyethylene and polypropylene, which serve as essential raw materials for plastic converters globally.¹ Third, the power and utilities segment provides the basic infrastructure and public utility systems necessary for the operation of the Rayong industrial ecosystem.¹ Fourth, the port and tank farm business provides international standard dock services for domestic and international cargo, facilitating the logistics of petroleum and petrochemical distribution.¹ Fifth, the asset management business focuses on the sale, lease, and granting of usage rights for land and buildings to support commercial activities aligned with the company's strategic direction.¹

IRPC's position within the energy sector is further defined by its majority ownership. As of 2024, PTT Public Company Limited (PTT Group) holds a 47.55% stake in the company, ensuring

that IRPC's operational and sustainability strategies are closely aligned with the broader mandates of Thailand's national energy leader.² The company's sector classification on the SET is Resources, specifically under the Energy & Utilities industry.¹ This identification is verified through official exchange factsheets and company investor relations disclosures.¹

Historically, IRPC began its journey in 1978 as Thai Petrochemical Industry Public Company Limited.² Following the 1997 Asian financial crisis, the company underwent a rigorous financial rehabilitation process that successfully concluded in 2006, at which point it adopted its current name and strategic focus.² By 2023, the organization employed approximately 5,325 full-time personnel and maintained a market capitalization exceeding 25 billion Thai Baht.² This historical resilience has informed its current ESG approach, characterized by a transition from financial recovery to long-term sustainability leadership.

ESG Report Timeline and Disclosure Evolution (2014–2023)

The decade from 2014 to 2023 represents the maturation of IRPC's sustainability reporting, moving from basic corporate social responsibility (CSR) summaries to high-density, integrated disclosures aligned with global financial and environmental standards.

The Benchmark Year: 2014 and Global Index Entry

The year 2014 served as a critical inflection point for IRPC's global ESG standing. For the first time, the company was selected for membership in the Dow Jones Sustainability Indices (DJSI) under the Emerging Markets category.⁸ This recognition was particularly significant as IRPC was one of only 14 companies selected from a global pool of 91 major oil and gas producers, signaling to international institutional investors that the company's governance and environmental management met rigorous global benchmarks.⁸ During this period, the company was also recognized in the RobecoSAM Sustainability Yearbook 2015, receiving the "Bronze Class" distinction and the "Industry Improver" award.⁸ These accolades were not merely symbolic; they represented a successful effort to resolve over 100 historical legal disputes with former shareholders, thereby stabilizing the governance pillar of the company's ESG profile.⁸

In terms of disclosure, the 2014 reporting period focused on the "Phoenix Project," a strategic initiative aimed at productivity enhancement and refinery modernization.⁸ The sustainability data for this year was reviewed by the Audit Committee and audited by external providers, ensuring that the integration of environmental and social KPIs began to mirror the rigor of financial reporting.⁹

Strategic Consolidation: 2015–2018

Following its entry into the DJSI, IRPC formalized its sustainability approach through the introduction of the "Sustainability Excellence Framework" in 2015.¹⁰ This framework was designed to embed environmental management policies directly into the operational heart of the refinery and petrochemical plants. The reporting during these years emphasized the "UHV Project" (Upgrading Heavy Feedstock to Value), which aimed to increase the yield of high-value refined products while reducing the production of low-value, high-emission fuel oil.¹¹

By 2016 and 2017, IRPC's disclosures began to more explicitly reference the UN Sustainable Development Goals (SDGs), mapping corporate activities to global targets such as Climate Action (SDG 13) and Responsible Consumption and Production (SDG 12).¹⁰ The company also maintained its inclusion in the FTSE4Good Emerging Index during this period, which reinforced its appeal to ESG-focused capital.¹² Third-party audits remained a staple of the disclosure process, with auditors such as EY and PwC providing assurance on different aspects of the financial and non-financial statements.⁴

Framework Alignment and the PTT Synergies: 2019–2021

In 2019, IRPC underwent a major revision of its Sustainability Excellence Framework to align more closely with the PTT Group Sustainability Management Framework.¹⁰ This move was a second-order strategic response to the increasing demand for standardized data across the PTT subsidiaries. The revised framework incorporated the "Sufficient Economy" philosophy alongside international standards, reflecting a blend of local Thai cultural values and global ESG expectations.¹⁰

The reporting years of 2020 and 2021 were defined by the organizational response to the COVID-19 pandemic and the acceleration of the energy transition strategy. IRPC's disclosures during this time began to incorporate more robust climate-related data, reflecting the early adoption of the Task Force on Climate-related Financial Disclosures (TCFD) recommendations.¹⁴ The company's environmental reporting through the Carbon Disclosure Project (CDP) gained prominence, as investors increasingly demanded transparency regarding water security and carbon intensity in the petrochemical industry.¹⁶

Integrated Value and Green Finance: 2022–2023

The culmination of IRPC's ESG evolution occurred in 2022 and 2023 with the full adoption of the "56-1 One Report" format, mandated by the Thai Securities and Exchange Commission (SEC).¹⁸ This report integrates financial, environmental, and social disclosures into a single, comprehensive document. In March 2022, IRPC launched its "Green Financing Framework," which received a second-party opinion from recognized experts and aligned with the ICMA Green Bond Principles and ASEAN Green Bond Standards.⁶ This framework allowed the company to issue green debentures specifically for projects like the 12.5 MW Floating Solar Farm and refinery efficiency upgrades to meet Euro 5 standards.⁶

By 2023, IRPC's reporting demonstrated a "Net Zero" ambition, setting targets for greenhouse gas (GHG) reduction by 20% by 2030 and reaching Net Zero by 2060.¹⁰ The 2023 report highlighted the successful launch of the first batch of Euro 5 diesel, a major regulatory and environmental milestone.¹⁸ This development was supported by a 5-billion-Baht sustainability loan from the Government Savings Bank (GSB), which explicitly used IRPC's ESG scores to determine lending eligibility and terms.²²

Year	Significant ESG Reporting Milestone	Frameworks/Indices Mentioned	Relevance to ESG Performance	Source
2014	First-time inclusion in DJSI Emerging Markets	DJSI, RobecoSAM	Established global sustainability credibility for the first time.	8
2015	Launch of Sustainability Excellence Framework	GRI, Internal Policy	Formalized the integration of ESG into business operations.	10
2019	Framework revision for PTT Group alignment	GRI Standards, SDGs	Harmonized ESG data across parent company and subsidiaries.	10
2020	Focus on Resilience and COVID-19 response	TCFD (initial), GRI	Demonstrated social pillar strength during global crisis.	23
2022	Launch of Green Financing	GBP, ASEAN GBS, GLP	Linked sustainability targets directly	6

	Framework		to capital market instruments.	
2023	Transition to Euro 5 and Net Zero Roadmap	GRI 2021, TCFD, SASB	Aligned refinery output with regional environmental regulations.	18

Adoption of International ESG Frameworks and Standards

The depth of IRPC's reporting is supported by its multifaceted adoption of international frameworks, each serving a specific stakeholder audience and disclosure purpose.

Global Reporting Initiative (GRI)

The Global Reporting Initiative (GRI) remains the primary standard for IRPC's sustainability disclosures. The analysis of the company's 2023 One Report confirms that it prepares disclosures "in accordance" with the GRI Standards 2021.¹⁰ This involves a rigorous materiality assessment process where IRPC identifies and prioritizes the ESG issues most relevant to its stakeholders, such as energy management, occupational health and safety, and business ethics.¹⁰ The use of GRI allows for comparability with global peers in the energy and chemical sectors, providing a standardized set of metrics for carbon emissions, water withdrawal, and waste management.

Task Force on Climate-related Financial Disclosures (TCFD)

To address the growing concerns of the financial community regarding climate risk, IRPC has integrated TCFD recommendations into its reporting structure.¹⁴ This framework focuses on four thematic areas: Governance, Strategy, Risk Management, and Metrics and Targets. IRPC utilizes TCFD to disclose how the transition to a low-carbon economy—including regulatory shifts like the EU's Carbon Border Adjustment Mechanism (CBAM)—could impact its financial performance and long-term asset value.²² The company's scenario analysis, which considers different temperature pathways, is a direct application of TCFD principles.¹⁴

Sustainability Accounting Standards Board (SASB)

Complementing the GRI's impact-focused reporting, IRPC utilizes SASB standards to provide financially material ESG information to investors.¹⁵ For an integrated refinery and petrochemical company, the SASB industry standards for "Chemicals" and "Oil & Gas – Refining & Marketing"

are particularly relevant. These metrics include the number of process safety incidents, the percentage of hazardous waste recycled, and the carbon footprint of the refining process.¹⁵ The integration of SASB helps analysts calculate the potential financial impact of ESG factors on the company's EBITDA and valuation multiples.¹

Carbon Disclosure Project (CDP)

IRPC's environmental transparency is further validated through its participation in the CDP.¹⁶ By disclosing to the CDP's Climate Change and Water questionnaires, the company provides granular data on its Scope 1, 2, and 3 emissions, as well as its water stewardship practices in the water-stressed region of Rayong.¹⁰ Within the PTT Group, IRPC benefits from shared best practices, as peer companies like PTT Global Chemical (GC) have achieved the prestigious "A" rating from CDP, serving as a benchmark for IRPC's internal performance improvements.²³

ESG Ratings and Independent Assessments

Third-party ratings serve as an external mirror for IRPC's internal ESG reporting, providing institutional investors with an objective measure of the company's risk profile and performance.

MSCI ESG Ratings

MSCI ESG Research evaluates IRPC on a scale from AAA (Leader) to CCC (Laggard).²⁸ The rating is determined by IRPC's exposure to industry-specific ESG risks and its ability to manage those risks relative to its peers. For a petrochemical entity, MSCI focuses on "Key Issues" such as toxic emissions and waste, carbon emissions, and corporate governance.²⁹ IRPC's efforts in developing eco-friendly products, such as Euro 5 diesel and recyclable polymers, serve as positive drivers for its MSCI rating.¹⁸ Historical data from 2022 and 2023 suggests that Thai energy companies in the PTT group typically fall into the "BBB" to "AA" range, reflecting a "Leader" or "Above Average" status within the emerging markets universe.³²

Morningstar Sustainalytics ESG Risk Ratings

Sustainalytics provides a "Risk Rating" score where a lower number indicates less unmanaged ESG risk.²⁹ The methodology distinguishes between "Manageable Risk" and "Unmanageable Risk." For IRPC, carbon emissions from refinery operations are considered a partially unmanageable risk due to the nature of the business model; however, the company's investments in energy efficiency and renewable energy (such as the floating solar farm) act as management scores that reduce the overall risk rating.⁶ Sustainalytics also monitors "Controversies," such as environmental accidents or safety lapses, which can negatively impact the final score.³⁵

FTSE4Good Index Series

IRPC has been a consistent constituent of the FTSE4Good Emerging Index.¹² This index requires

companies to meet specific ESG criteria across various pillars, including climate change mitigation, labor standards, and anti-corruption.¹³ Inclusion in this index is a prerequisite for many sustainable investment funds and demonstrates that IRPC's practices align with the values of global ethical investors.¹²

SET ESG Ratings (Thailand Sustainability Investment - THSI)

On the local exchange, IRPC has consistently achieved top-tier ratings in the SET ESG Ratings (formerly the Thailand Sustainability Investment or THSI list).¹ In 2023, the company received an "AAA" rating, the highest possible score, reflecting its comprehensive disclosure and performance in the Thai context.¹ This rating is highly valued by domestic institutional investors and Thai pension funds.³³

Financial Integration of ESG: Green Financing and Sustainability Loans

One of the most significant developments in IRPC's ESG strategy between 2014 and 2023 is the direct link established between sustainability performance and financial strategy.

The Green Financing Framework

In March 2022, IRPC established its "Green Financing Framework," a document that outlines the criteria for projects eligible for funding through green bonds or green loans.⁶ This framework is built on four pillars: Use of Proceeds, Process for Project Evaluation and Selection, Management of Proceeds, and Reporting.⁶ Projects eligible for green financing include renewable energy (such as floating and rooftop solar), energy efficiency improvements (refinery upgrades), and waste-to-value initiatives (circular economy projects).⁶

Green Debentures and Allocation of Proceeds

Following the launch of the framework, IRPC successfully issued green debentures to institutional and retail investors. According to the "IRPC Green Annual Report 2024," the company has fully allocated the proceeds as of December 31, 2023.⁶ Approximately 71.77% of the funds were used for refinancing existing sustainable projects, while 28.23% supported new financing for green initiatives.⁶ A key beneficiary of this funding was the Floating Solar Farm project, which not only provides clean energy to the refinery but also reduces water evaporation from the storage reservoirs, demonstrating a "Water-Energy Nexus" insight.⁶

Sustainability-Linked Loans (SLL)

In 2023, IRPC secured a 5-billion-Baht sustainability-linked loan from the Government Savings Bank (GSB).²² Unlike traditional green bonds which are project-specific, this loan's terms are linked to IRPC's overall ESG scores. This indicates a high level of confidence from the banking

sector in IRPC's sustainability management. The loan was utilized to fund the Ultra Clean Fuel (UCF) project, specifically the production of Euro 5 standard diesel, which reduces particulate matter (PM 2.5) and other harmful emissions.²²

Innovative Financial Derivatives

IRPC has also collaborated with Siam Commercial Bank (SCB) to initiate "Sustainability-Linked Forward Contracts".²² This financial derivative innovation allows the company to hedge foreign exchange risks while incentivizing sustainable development. If the company meets its sustainability targets, it receives more favorable terms on its forward contracts, effectively lowering its financial risk through ESG performance.²²

Operational ESG Projects and Material Milestones

The narrative of IRPC's ESG journey is best illustrated through the concrete projects and operational milestones achieved during the 2014–2023 period.

The Euro 5 Diesel Transition

The launch of Euro 5 diesel in 2023 was a landmark achievement for IRPC's environmental strategy.¹⁸ This project, known as the Ultra Clean Fuel (UCF) project, involved a multi-billion Baht investment in desulfurization technology at the Rayong refinery.²² The transition from Euro 4 to Euro 5 involves a significant reduction in sulfur content from 50 parts per million (ppm) to 10 ppm.¹⁸ This second-order insight reveals that IRPC is not just meeting regulation but is actively contributing to the solution for Thailand's seasonal haze and air quality issues, thereby enhancing its "Social License to Operate".¹⁸

Renewable Energy and the Floating Solar Farm

IRPC has utilized its industrial water reservoirs to host a 12.5 MW Floating Solar Farm, which was expanded by an additional 8.5 MW in 2023.⁶ This project serves multiple ESG goals: it reduces the refinery's dependence on fossil-fuel-based grid electricity, lowers Scope 2 emissions, and optimizes the use of industrial land.⁶ The project is expected to reduce GHG emissions by approximately 9,459 tons of CO₂ equivalent per year.¹⁰

The Circular Economy and the BCG Model

Aligned with the Thai government's Bio-Circular-Green (BCG) model, IRPC has implemented several circular economy initiatives.¹⁸ The "KFC Recycled Plastic" project is a prime example, where IRPC collaborated with the fast-food chain to collect and recycle plastic waste into new products, such as trays and furniture for branches.¹⁰ Furthermore, the company's development of "REINFOXX" fertilizer—a nutrient-rich product derived from byproduct streams—demonstrates how the company is turning waste into social and economic value for

the agricultural sector.¹⁸

Reforestation and Carbon Sequestration

IRPC has partnered with the Royal Forest Department and the SET for the "Care the Wild" project, focusing on reforestation in the Ban Khok Phluang community forest.¹⁸ This 10-year commitment involves not just planting trees but ensuring their survival to maximize carbon sequestration.¹⁸ This project reflects IRPC's broader target of accelerating planting in additional forest areas as part of its Net Zero roadmap.¹⁸

Governance and Ethics: The Foundation of ESG

The "Governance" pillar at IRPC is defined by its anti-corruption stance, board diversity, and rigorous audit processes.

Anti-Corruption and Transparency

IRPC is a certified member of the Thai Private Sector Collective Action Against Corruption (CAC).¹ This certification requires a comprehensive audit of internal policies, gift policies, and whistleblower mechanisms. The company's "Excellent" rating in the Corporate Governance (CG) Report of Thai Listed Companies further underscores its commitment to transparency.¹

Board of Directors and Audit Committee

The governance structure includes an Audit Committee composed entirely of independent directors.¹ This committee is responsible for overseeing the integrity of financial and sustainability disclosures. In 2014, for instance, the Audit Committee played a vital role in resolving historical legal disputes, which had been a significant governance overhang.⁸ The board also includes specialized committees for Corporate Governance and Sustainability to ensure that ESG strategy is discussed at the highest level of leadership.⁸

External Audits and Assurance

A critical requirement for ESG credibility is external assurance. IRPC's reports have historically been audited by major firms such as EY and PwC.⁴ These audits cover not just the financial statements but also specific ESG indicators, such as energy consumption and safety rates, as required by the GRI Standards.¹⁰ The use of "Limited Assurance" for sustainability data is a standard practice that provides stakeholders with confidence in the accuracy of the reported metrics.²⁰

Reference Table and Credibility Assessment

The following table presents the key sources used for this report, evaluated for their credibility

and relevance according to the specified guidelines.

#	Reference (APA 7th edition format)	Source Type	Credibility Score (0–10)	Justification for Score	Relevance to Query
1	IRPC Public Company Limited. (2024). <i>Annual Report 2023 (Form 56-1 One Report)</i> .	Primary (Official filing)	10	Mandated regulatory disclosure, audited by external professionals.	High: Contains the most recent ESG milestones and targets.
2	Stock Exchange of Thailand. (2026). <i>IRPC Public Company Limited Factsheet</i> .	Primary (Exchange data)	10	Official market data from the listing exchange; high transparency and recency.	High: Confirms legal name, HQ, and industry classification.
3	IRPC Public Company Limited. (2015). <i>Annual Report 2014</i> .	Primary (Company Report)	10	Primary historical record; audited financial and CSR data.	High: Key for DJSI entry and historical baseline analysis.
4	Sustainalytics . (2023). <i>ESG Risk Ratings</i>	Reputable Secondary	8	Globally recognized rating	Medium: Explains external ESG

	and Methodology .			provider; transparent methodology but proprietary data.	risk assessment.
5	IRPC Public Company Limited. (2022). <i>IRPC Green Financing Framework</i> .	Primary (Policy)	9	Detailed framework verified by second-party opinions; foundational for green debt.	High: Explains the link between ESG and finance.
6	CDP. (2023). <i>Global Environmental Disclosure Reports</i> .	Reputable Secondary	8	Standard-sett ing NGO for climate and water data; used by thousands of companies.	Medium: Provides context for environmental disclosure.
7	Reuters/Bloo mberg Financial Data (via Factsheets). (2026). <i>IRPC Ticker Data</i> .	Reputable Secondary	9	Real-time financial data with strict editorial and reporting standards.	High: Used for identification and market stats.

Insights, Trends, and Future Implications

The synthesis of IRPC's ESG performance over the last decade reveals several underlying trends that indicate the future trajectory of the company and the broader energy sector in

Thailand.

From Compliance to Competitive Advantage

Between 2014 and 2018, ESG at IRPC was largely a matter of "compliance and recognition," focused on getting into indices like the DJSI.⁸ However, the 2019–2023 period shows a shift toward using ESG as a "competitive advantage." The second-order insight here is that IRPC's high ESG ratings allowed it to access cheaper capital via sustainability-linked loans and green bonds.⁶ In an environment of rising interest rates, having an "ESG edge" becomes a financial tool for lowering the cost of debt.

The Decarbonization Imperative and Asset Stranding

The transition to Euro 5 diesel and the Net Zero 2060 roadmap are responses to a fundamental risk: the potential for "stranded assets" in a post-fossil-fuel world.¹⁰ By diversifying into specialty petrochemicals, floating solar, and recycled polymers, IRPC is proactively hedging against the decline of traditional fuel demand.⁶ The causal relationship between EU environmental regulations (like CBAM) and IRPC's internal carbon tracking projects shows that regional players must adapt to global carbon pricing mechanisms even if they primarily serve domestic markets.²²

The PTT Ecosystem Multiplier

IRPC's affiliation with the PTT Group (holding 47.55%) acts as a sustainability multiplier.² By aligning with PTT's broader Net Zero targets, IRPC benefits from shared technology, joint ventures in renewable energy, and a unified ESG reporting language.¹⁰ This ecosystem approach reduces the individual cost of ESG transition for IRPC, providing a safety net of technical and financial expertise.

Social Pillar and Community Resilience

The focus on community forests and vocational education (IRPC Technological College) indicates that the "Social" pillar is being treated as a long-term investment in community resilience.¹⁸ As the petrochemical complex is located in Rayong, maintain harmony with the local community is essential to prevent operational disruptions and legal disputes, similar to those resolved in 2014.⁸

Transparency and Data Limitations

While every effort has been made to provide a comprehensive and verified analysis, certain limitations exist in the available research material.

Knowledge Cutoff and Real-Time Data

The information presented reflects data available up to March 20, 2026.¹ Any strategic shifts, financial results, or environmental incidents occurring after this date are not included. Furthermore, while the company's "One Report" is highly detailed, some proprietary data regarding specific product-level life cycle assessments (LCAs) or detailed Scope 3 emission calculations per supplier are often not publicly disclosed in full detail.

Verification of Third-Party Ratings

Specific historical letter ratings from MSCI or numerical scores from Sustainalytics for each year from 2014 to 2023 are sometimes proprietary and may vary slightly between different data providers (e.g., Bloomberg vs. Refinitiv). This report has utilized general performance trends and official inclusions in public indices (DJSI, FTSE4Good, SET ESG) as verified proxies for these ratings.¹²

Unverified Claims

Claims regarding the "first world-standard coatings" or specific carbon reduction percentages for upcoming projects in 2024–2025 are labeled as "forward-looking" and remain unverified until officially audited in future annual reports.¹⁰ The feasibility of reaching Net Zero by 2060 remains a strategic ambition rather than a guaranteed fact, subject to future technological and policy changes.

Recommended External Databases

For users seeking to independently verify or deep-dive into IRPC's ESG data, the following databases are recommended:

1. **SET ESG Ratings Portal:** Specifically for Thai market comparisons and THSI historical inclusions.
2. **GRI Standards Database:** To search for IRPC's official GRI report filings and compliance levels.
3. **CDP Disclosure Platform:** For detailed climate change and water security responses submitted by IRPC and its parent, PTT Group.
4. **IRPC Investor Relations Portal:** For the full archive of 56-1 One Reports and Green Financing Framework documents.⁷

Conclusion: A Trajectory of Sustainable Transformation

The detailed analysis of IRPC Public Company Limited between 2014 and 2023 reveals a company that has successfully moved from the periphery to the center of the global sustainability conversation. Starting with its benchmark entry into the DJSI in 2014, IRPC has systematically built a foundation of transparency, governance, and operational excellence.⁸ The adoption of international frameworks such as GRI, TCFD, and SASB has not only satisfied

regulatory requirements but has also facilitated a new era of green financing, where the company's ESG performance directly impacts its financial health.⁶

The operational transition to Euro 5 diesel and the investment in renewable energy through floating solar farms are tangible markers of a strategy designed to survive and thrive in a low-carbon future.⁶ While the challenges of operating in a carbon-intensive industry remain significant, IRPC's integrated approach—combining industrial prowess with environmental stewardship and social responsibility—positions it as a resilient leader in the Southeast Asian energy sector. The next decade will likely be defined by the company's ability to execute its Net Zero roadmap and continue its evolution toward a circular, bio-based economy.

Trabalhos citados

1. IRPC - Factsheet - The Stock Exchange of Thailand - SET, acesso a março 19, 2026, <https://www.set.or.th/en/market/product/stock/quote/irpc/factsheet>
2. IRPC.BK Price: Quote, Forecast, Charts & News - Perplexity, acesso a março 19, 2026, <https://www.perplexity.ai/finance/IRPC.BK/earnings>
3. IRPC Public Company Limited Price: Quote, Forecast, Charts & News (IRPC.BK) - Perplexity, acesso a março 19, 2026, <https://www.perplexity.ai/finance/IRPC.BK>
4. IRPC - Company profile - The Stock Exchange of Thailand - SET, acesso a março 19, 2026, <https://www.set.or.th/en/market/product/stock/quote/IRPC/company-profile/information>
5. IRPC PCL, IRPC:SET summary - FT.com - Markets data, acesso a março 19, 2026, <https://markets.ft.com/data/equities/tearsheet/summary?s=IRPC:SET>
6. IRPC GREEN ANNUAL REPORT 2024, acesso a março 19, 2026, <https://investor.irpc.co.th/storage/content/downloads/debenture-holders-information/20250221-irpc-green-debenture-report-en.pdf>
7. Financial Statements | IRPC Public Company Limited, acesso a março 19, 2026, <https://investor.irpc.co.th/en/downloads/financial-statements?year=2023>
8. IRPC 2014 Eng_A-Coated Toyo.ai-Cs6-Mac08, acesso a março 19, 2026, <https://irpc.listedcompany.com/misc/ar/20150303-irpc-ar-2014-en.pdf>
9. Invitation to attend the 2014 Annual General Meeting of Shareholders IRPC Public Company Limited, acesso a março 19, 2026, <https://investor.irpc.co.th/storage/content/downloads/shareholders-meeting/agm2014/20140409-irpc-agm2014-invitation-full-en-03.pdf>
10. IRPC Green Financing Framework: March 2022 - ThaiBMA, acesso a março 19, 2026, [https://www.thaibma.or.th/Download/prospetus/IRPC325A/\(6\)150801_25650509.pdf](https://www.thaibma.or.th/Download/prospetus/IRPC325A/(6)150801_25650509.pdf)
11. minutes of the 2014 annual general meeting of shareholders - IRPC, acesso a março 19, 2026, <https://investor.irpc.co.th/storage/content/downloads/shareholders-meeting/agm2014/20140515-irpc-agm2014-minutes-en-02.pdf>
12. FTSE4Good Index Series - LSEG, acesso a março 19, 2026,

- <https://www.lseg.com/en/ftse-russell/indices/ftse4good>
13. FTSE4Good Index Series - FTSE Russell Research Portal, acesso a março 19, 2026, <https://research.ftserussell.com/Analytics/FactSheets/Home/DownloadSingleIssue?issueName=4GL1&isManual=False>
 14. The Evolution of Sustainability Reporting: Trends from GRI to ISSB - The Overview Effect, acesso a março 19, 2026, <https://theovervieweffect.nl/insights/the-evolution-of-sustainability-reporting/>
 15. Introduction to Popular ESG Reporting Frameworks Like GRI, SASB, and TCFD - eSystems, acesso a março 19, 2026, <https://www.esystems.fi/en/blog/introduction-to-popular-esg-reporting-frameworks-like-gri-sasb-and-tcfd>
 16. Record 23000+ companies disclose environmental impact through CDP, with urgency for action clear in wake of unprecedented global temperatures, acesso a março 19, 2026, <https://www.cdp.net/en/press-releases/record-23-000-companies-disclose-environmental-impact-through-cdp-with-urgency-for-action-clear-in-wake-of-unprecedented-global-temperatures>
 17. CDP Scores 2023 - DFGE - Institute for Energy, Ecology and Economy, acesso a março 19, 2026, <https://dfge.de/cdp-scores-2023/>
 18. Annual Report 2023 (Form 56-1 One Report) IRPC Public Company Limited, acesso a março 19, 2026, <https://irpc.listedcompany.com/misc/or/20240307-irpc-or2023-en.pdf>
 19. The History of Integrated Reporting - Report Yak, acesso a março 19, 2026, <https://reportyak.com/blog/a-brief-history-of-integrated-reporting/>
 20. GPSC SD2023, acesso a março 19, 2026, <https://www.gpscgroup.com/storage/download/sd-report/20240409-gpsc-sd2023-en.pdf>
 21. Sustainability Highlight 2023 - SET, acesso a março 19, 2026, https://media.set.or.th/set/Documents/2024/Mar/sd_report_2023_en.pdf
 22. Key Performance - IRPC, acesso a março 19, 2026, <https://irpc.listedcompany.com/misc/or/20250305-irpc-or2024-en.pdf>
 23. BETTER LIVING THROUGH RESILIENCE - Sustainability | PTT Global Chemical, acesso a março 19, 2026, <https://sustainability.pttggroup.com/storage/document/annual-report/2020/20210308-pttgc-one-report2020-en.pdf>
 24. GPSC IFRS S2 Sustainability Disclosure Standard 2024, acesso a março 19, 2026, <https://www.gpscgroup.com/storage/download/sustainability/environmental/climate-change-management/20240628-gpsc-ifrs-s2-sustainability-disclosure-standard-2024-th.pdf>
 25. Five global organisations, whose frameworks, standards and platforms guide the majority of sustainability and integrated reporting, announce a shared vision of what is needed for progress towards comprehensive corporate reporting - CDP, acesso a março 19, 2026, <https://www.cdp.net/en/press-releases/comprehensive-corporate-reporting>
 26. CDP vs SASB vs GRI Sustainability Reporting Standards | Blog - OneTrust, acesso

- a março 19, 2026,
<https://www.onetrust.com/blog/cdp-sasb-gri-sustainability-reporting-standards/>
27. CDP Scores and A Lists, acesso a março 19, 2026,
<https://www.cdp.net/en/data/scores>
 28. Top ESG Ratings Providers: Ecovadis, CDP Questionnaire, S&P Global ESG, MSCI ESG Research, Sustainalytics, FTSE4Good and JUST Capital -
www.positiveorg.co, acesso a março 19, 2026,
<https://www.positiveorg.co/en/news-publications/top-esg-ratings-providers-ecovadis-cdp-questionnaire-s-p-global-esg-msci-esg-research-sustainalytics-ftse4-good-and-just-capital>
 29. ESG Scores Explained: MSCI vs Sustainalytics vs S&P Global Ratings, acesso a março 19, 2026,
<https://www.futuregreenworld.com/post/esg-scores-explained-msci-vs-sustainalytics-vs-s-p-global-ratings>
 30. ESG Ratings Assess companies on their financially relevant sustainability risks and opportunities. - MSCI, acesso a março 19, 2026,
<https://www.msci.com/data-and-analytics/sustainability-solutions/esg-ratings>
 31. ESG Ratings Model History - MSCI, acesso a março 19, 2026,
<https://www.msci.com/documents/1296102/34424357/MSCI+ESG+Ratings+Model+Update+History.pdf/6155b263-2bf7-1c01-cf49-6d74c9b927e6?t=1713205717580>
 32. Minor International MINT - InnovestX, acesso a março 19, 2026,
https://www.innovestx.co.th/docs/default-source/about-us/mint230704_e2fbf7aa7-6bc3-4305-83e4-88450e110e5a.pdf?sfvrsn=d32f799_1
 33. EMPOWERING THE WORLD COMPASSIONATELY - B.Grimm Power Public Company Limited, acesso a março 19, 2026,
<https://www.bgrimpower.com/storage/download/sustainability-report/2024/20250326-bgrim-one-report-2024-en.pdf>
 34. EMPOWERING THE WORLD COMPASSIONATELY - B.Grimm Power Public Company Limited, acesso a março 19, 2026,
<https://bgrim.listedcompany.com/misc/one-report/20240322-bgrim-one-report-2023-en.pdf>
 35. ESG Risk Ratings - Sustainalytics, acesso a março 19, 2026,
<https://www.sustainalytics.com/esg-data>
 36. ESG Risk Ratings - Sustainalytics, acesso a março 19, 2026,
<https://www.sustainalytics.com/corporate-solutions/esg-solutions/esg-risk-ratings>
 37. THE ESG RISK RATINGS - Sustainalytics, acesso a março 19, 2026,
https://www.sustainalytics.com/docs/default-source/meis/mei-campaign_resilience-backgrounder_final-2.pdf?sfvrsn=6a749f3_0
 38. FTSE4Good Index Series Ground Rules - LSEG, acesso a março 19, 2026,
https://www.lseg.com/content/dam/ftse-russell/en_us/documents/ground-rules/ftse4good-index-series-ground-rules.pdf
 39. FTSE4Good Index - Wikipedia, acesso a março 19, 2026,
https://en.wikipedia.org/wiki/FTSE4Good_Index
 40. 56-1 One Report - AMATA, acesso a março 19, 2026,

<https://amata.com/documents/20240328-amata-one-report2023-en.pdf>

41. pttep-sustainability-report-2014-en.pdf, acesso a março 19, 2026,
<https://hub.optiwise.io/storage/129/sustainability-report/2014/pttep-sustainability-report-2014-en.pdf>
42. Dow Jones Sustainability Indices - Wikipedia, acesso a março 19, 2026,
https://en.wikipedia.org/wiki/Dow_Jones_Sustainability_Indices